

Introduction to Dose-Response Physical Activity (drpa) package

Last updated on: 16 September, 2026

Installation from GitHub

The package is not available at CRAN. It needs to be installed from GitHub. Please use this command to install it.

```
remotes::install_github("meta-analyses/drpa")
```

You may need to install the `remotes` package first.

List of diseases/cause and their upper limit

The first column `disease` specifies the supported diseases in the package. The other three columns, which are outcome types (`fatal`, `non-fatal` and `fatal-and-non-fatal`), show the last knot (based on the available data points) for the specific disease. We don't have confidence that our meta-analysis is reliable after this point, and assumes an onward flat relationship. Whenever a disease such as `all-cause-cvd` doesn't have an outcome type `non-fatal`, it will be NA. More will follow in the next section.

Setup

Load the package

```
library(drpa)
```

Get dose response relationship

Get response for dose *40* for *all-cause-cvd* with the default quantile *0.5* and censoring at doubling of WHO's recommended level of physical activity for adults (*17.5* MMETs hours per week).

The `quantile`'s value is between 0 and 1 [excluding the boundary values]

```
drpa::dose_response(cause = 'all-cause-cvd', outcome_type = 'fatal-and-non-fatal', dose = 40,
                    quantile = 0.5, confidence_intervals = T, censor_method = 'WHO-DRL')
#>           rr           lb           ub
#> 1 0.6736267 0.6273602 0.7233052
```

disease	fatal	non-fatal	fatal-and-non-fatal
all-cause-cancer	19.13777	NA	19.49427
all-cause-cvd	21.33552	NA	20.11288
all-cause-dementia	NA	13.51200	11.41597
all-cause-mortality	21.86500	NA	NA
alzheimers-disease	NA	22.36750	13.63786
bladder-cancer	NA	NA	13.45000
breast-cancer	15.47684	NA	26.75480
colon-cancer	NA	NA	20.10136
coronary-heart-disease	11.00136	NA	17.96000
depression	NA	14.56452	14.56452
depressive-symptoms	NA	11.75641	11.75641
diabetes	NA	NA	NA
endometrial-cancer	NA	NA	21.16857
esophageal-cancer	NA	NA	21.82207
gastric-cardia-cancer	NA	NA	22.52300
head-and-neck-cancer	NA	NA	13.59235
heart-failure	NA	NA	24.41750
kidney-cancer	NA	NA	14.81429
liver-cancer	NA	NA	21.82154
lung-cancer	20.20636	NA	14.17460
major-depression	NA	16.54304	16.54304
myeloid-leukemia	NA	NA	10.06429
myeloma	NA	NA	21.83250
parkinsons-disease	NA	15.11423	15.17000
prostate-cancer	14.87857	NA	27.26046
rectum-cancer	NA	NA	18.09230
stroke	11.00400	NA	15.01652
vascular-dementia	NA	NA	16.85333

Arguments

Description of the arguments are mentioned below:

Argument	Description
cause	All-cause-mortality or disease name
outcome_type	Either one of the three possibilities for outcome type: fatal, non-fatal or fatal-and-non-fatal
dose	Dose (all-cause or disease)
quantile	Numeric value between 0 and 1 - default is 0.5. This captures the range of dose values between lower and upper confidence levels.
censor_method	This determines whether any censor method applies to the dose-response relationship. There are four possibilities: none , default (to use the default cut-off, determined by the meta-analysis), WHO-DRL (WHO's recommended physical activity per week 8.75 when doubled, so 17.5 MMETs hours per week) or WHO-QRL (WHO's recommended physical activity per week 8.75 when quadrupled, so 35 MMETs hours per week) . The default value is default - a fixed censor point at determined by the underlying meta-analysis after which the relationship remains the same.
confidence_intervals	Boolean variable to determine whether confidence intervals are returned or not - default false